

Networks Lab Assignment 2 Report

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1. Client and Server Terminals

Scenario 1: File Not Available in the Local Directory

Below are the terminal screenshots for the client and server when the file was not available in the local directory:

```
user@swlab-PC: ~/22CS10020$ ./client
Enter the file name: abc.txt
Sent file request for: abc.txt
FILE NOT FOUND
```

(a) Client terminal output.

```
user@swlab-PC: ~/22CS10020$ ./server
Server running...
Received file request for: abc.txt
File not found. NOTFOUND message sent to client.
```

(b) Server terminal output.

Figure 1: Client and server terminals when the file was not available.

Scenario 2: File Available in the Local Directory

Below are the terminal screenshots for the client and server when the file was available in the local directory:

```
user@swlab-PC: ~/22CS10020$ ./client
Enter the file name: 22CS10020_file1.txt
Sent file request for: 22CS10020_file1.txt
Received word: CS61060
Received word: CS31029
Received word: CS11023
Received word: FINISH
Client terminated. Received file saved as received_file.txt
```

(a) Client terminal output.

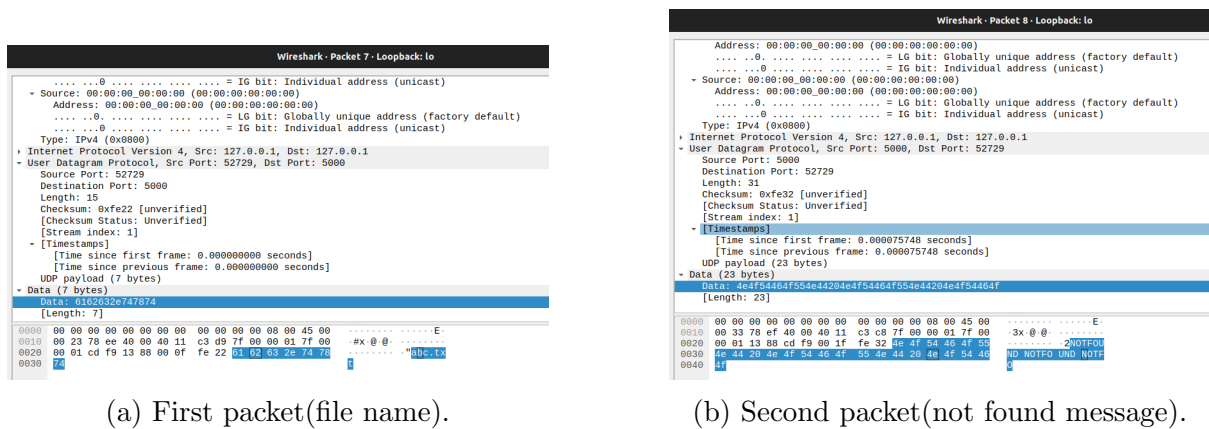
```
user@swlab-PC: ~/22CS10020$ ./server
Server running...
Received file request for: 22CS10020_file1.txt
Sent HELLO message to client.
Sent word: CS61060
Sent word: CS31029
Sent word: CS11023
Sent word: FINISH
Server terminated.
```

(b) Server terminal output.

Figure 2: Client and server terminals when the file was available.

2. Packet Exchange Screenshots

Scenario 1: File Not Available in the Local Directory



Scenario 2: File Available in the Local Directory

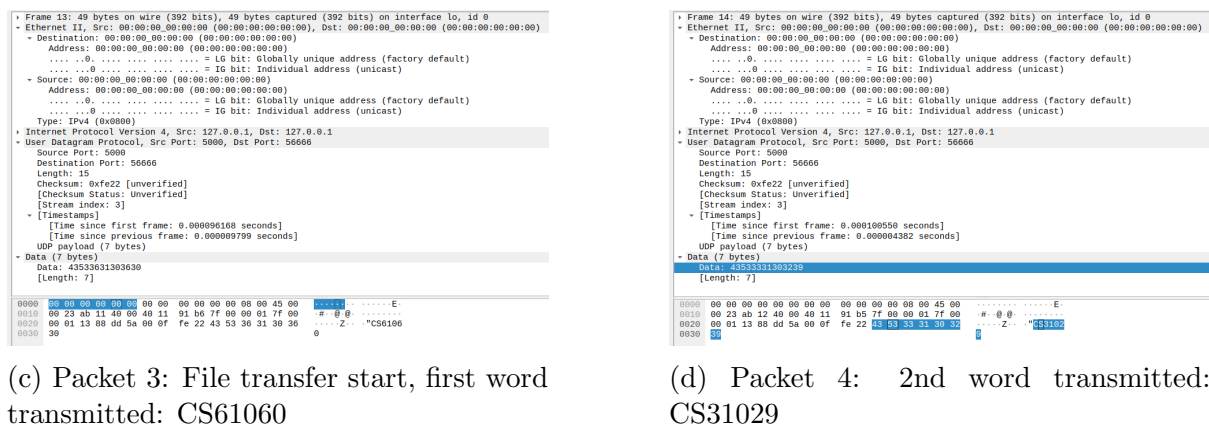
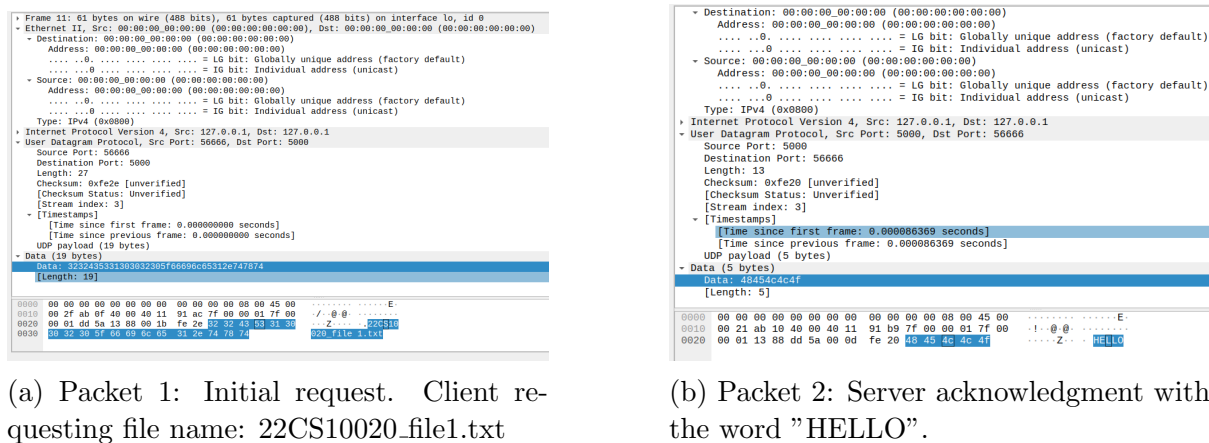


Figure 4: First four packets exchanged when the file was available.


```

> User Datagram Protocol, Src Port: 5000, Dst Port: 56666
  Source Port: 5000
  Destination Port: 56666
  Length: 13
  Checksum: 0xfe20 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 3]
  [Timestamps]
    [Time since first frame: 0.000086369 seconds]
    [Time since previous frame: 0.000086369 seconds]
  UDP payload (5 bytes)
    Data: 48454c4cf
    [Length: 5]

0000  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  08 0c 45 00  . . . . . E-
0010  00 21 ab 10 48 00 40 01 11 91 b9 7f 00 00 61 7f 00  ! . @ . . . .
0020  00 01 13 88 dd 5a 00 0d fe 29 48 45 4c 4c 4f 00  . . . . . Z . HELLO

```

```

User Datagram Protocol, Src Port: 5000, Dst Port: 56666
Source Port: 5000
Destination Port: 56666
Length: 15
Checksum: 0xfe22 [unverified]
[Checksum Status: Unverified]
[Stream index: 3]
  [Timestamps]
    [Time since first frame: 0.00009618 seconds]
    [Time since previous frame: 0.000009799 seconds]
  UDP payload (7 bytes)
    Data (7 bytes)
      Data: 43533631303630
      Length: 7

```

```
User Datagram Protocol, Src Port: 5000, Dst Port: 56666
Source Port: 5000
Destination Port: 56666
Length: 15
Checksum: 0xfe22 [unverified]
[Checksum Status: Unverified]
[Stream index: 3]
- [Timestamps]
    [Time since first frame: 0.000108559 seconds]
    [Time since previous frame: 0.000094382 seconds]
UDP payload (7 bytes)
- Data (7 bytes)
    Data: 43533331303239
    [Length: 7]
```

```
0000  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .E-
0010  ab 12 40 0d 00 00 00 00 11 91 b5 7f 00 00 81 7f 0e  # @ @ @
0020  00 01 13 88 dd 5a 00 0f fe 22 43 53 33 31 30 32   :Z:"CS3102
0030  39                                     9
```

[illegible]

```

> User Datagram Protocol, Src Port: 5000, Dst Port: 56666
  Source Port: 5000
  Destination Port: 56666
  Length: 14
  Checksum: 0xfe21 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 3]
  [Timestamps]
    [Time since first frame: 0.000126966 seconds]
    [Time since previous frame: 0.000022683 seconds]
  UDP payload (6 bytes)
  Data (6 bytes)
    Data: 46494ea95348
    [Length: 6]

```

The image shows a Wireshark packet capture of a UDP packet. The packet details pane on the left shows the following structure:

- User Datagram Protocol, Src Port: 5000, Dst Port: 56666
 - Source Port: 5000
 - Destination Port: 56666
 - Length: 14
 - Checksum: 0xfe21 [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 3]
 - [Timestamps]
 - [Time since first frame: 0.000126966 seconds]
 - [Time since previous frame: 0.000022683 seconds]
 - UDP payload (6 bytes)
 - Data (6 bytes)
 - Data: 46494ea95348
 - [Length: 6]

The packet bytes pane on the right shows the raw data of the packet. The data field (0000 to 0020) is highlighted in blue, corresponding to the data field in the packet details pane. The data is shown in hexadecimal and ASCII. The ASCII column shows the string "46494ea95348" followed by "FINISH".

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